

Enhancing Urban Transportation Resilience Using Graph Neural Networks with Social Event Integration

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Abstract—Urban Traffic Resilience Through Graph Neural Networks and Social Event Integration. Urban traffic resilience plays a vital role in daily transportation systems. Cities face increasing challenges from disruptions such as traffic congestion caused by construction, accidents, and major public events. To enable effective urban mobility planning, it is essential to forecast these disruptions and assess how quickly transportation systems can recover. A novel approach to predicting urban traffic resilience is introduced using Graph Neural Networks (GNNs) combined with integrated social event data. Urban traffic networks are modeled as dynamic graphs, which serve as simplified representations of traffic infrastructure. In these graphs, nodes represent critical locations such as intersections and roads, while edges indicate connections, including road links and traffic flow. Real-time data from sources such as mainstream media, social media platforms, and event planning systems is incorporated. This allows for consideration of disruptions caused by public events including concerts, protests, and sports gatherings, all of which have significant effects on traffic conditions. This method enhances the prediction of traffic congestion, recovery durations, and system adaptability. An advanced tool for urban traffic management is thereby provided. The model has been evaluated using real-world datasets, demonstrating its effectiveness in forecasting disruptions and offering strategies for improving traffic flow and resilience.

Index Terms—Urban Transportation Resilience, Graph Neural Networks (GNNs), Traffic Prediction, Social Event Data, Smart Cities, Spatio-Temporal Analysis, Machine Learning, Real-Time Traffic Management.

I. INTRODUCTION

Urban transportation networks are the pillar of contemporary cities, facilitating the uninterrupted movement of goods and people. Urban transportation networks are perennially disrupted by various factors such as traffic jams, road crashes, adverse weather, and large-scale social gatherings. Disruptions have the potential to significantly contribute to mobility by causing congestion, incurring economic losses, and affect quality of life for citizens. To improve the resilience of transportation networks, it is crucial to develop predictive models that can anticipate disruptions and enable proactive traffic management strategies. Recent advancements in Graph Neural Networks (GNNs) have enabled the modelling of complex, dynamic networks, making them suitable for analysing urban transportation systems. Unlike traditional machine learning models, GNNs are able to effectively learn traffic networks' inherent spatial and temporal dependencies, enabling more accurate prediction of incidents and recovery behaviours. By modelling transport networks as graphs, where nodes represent significant points (e.g., intersections, transit stations) and edges the relationships between them (e.g., roads, transit routes), GNNs are able to handle massive amounts of traffic data and identify patterns influencing urban mobility. Apart from natural traffic flows that happen on a daily basis, social events like concerts, sporting events, and protests also play a major role in causing traffic disruptions in cities. Such large groups of people, which in turn tend to draw large crowds, have the

potential to create high road congestion, patterns of traffic movement change, and high demands for public transport services. It should be noted, however, that most conventional traffic forecasting models do not account for the impacts of such social events. By utilizing real-time social event data from social media platforms, event websites, and public records, it is possible to significantly improve the predictability of traffic flows. This integration provides a more integrated and comprehensive approach to modelling and enhancing city traffic resilience, which enables better preparedness and traffic flow management during large social events.

This paper is centred on developing a predictive model from social event data and Graph Neural Network to enhance the resilience of city transport. The primary goals of this paper are:

1. City transport network modelling with GNNs for forecasting and analysis of traffic disruption.
2. Using real-time social event data to analyse big event impacts on urban traffic patterns.
3. Measuring the resilience of city transport systems, estimating the ability of networks to bounce back from disruptions.
4. Suggesting adaptive traffic management schemes using predictive analysis to enhance mobility in the city.

The framework shall be validated using real-world data sets, including traffic flows in a city and social event data, to show how it predicts disruption and enhances the resilience of transport networks. The work promotes the design of smart city transport systems via the delivery of data-led support to city planners, traffic administrators, and decision-makers.

II. LITERATURE REVIEW

A. Introduction

Urban transportation systems are complex networks that is the backbone in the mobility of people and goods in a particular city. However, these systems face significant challenges due to blockage, infrastructure limitations, and unpredictable disruptions such as accidents, climatic conditions, and large-scale social events and gatherings which occur spontaneously (Zheng et al., 2021). To enhance transportation resilience, it is crucial to develop predictive models that can predict and erase disruptions.

Graph Neural Networks (GNNs) have emerged as a powerful tool for modelling transportation systems due to their ability to process graph-structured data, capturing both spatial and temporal dependencies (Wu et al., 2020). Additionally, social event data from platforms like Twitter, Google Trends, and event organizers provide critical insights into traffic disturbances caused by large gatherings (Xu et al., 2022). This literature review explores the integration of GNNs with social event data to improve urban transportation resilience.

B. Resilience in Urban Transportation Systems

1) Robustness to the instant disruptions:

- Urban transportation systems often face unforeseen disruptions, such as accidents, road closures, or public events (concerts, festivals, protests). These

disruptions can lead to traffic congestion all over the city which is main reason of the delays.

- A resilient system is one that can flexibly manage these disturbances without completely breaking down, by redirecting the traffic which can be attained through deploying alternate routes, or efficiently utilizing available infrastructure.

2) Recovery Time of resilience system:

- A resilient transportation system doesn't just minimize disturbances; it also adapts the disruptions and recovers quickly from them.
- for example, if a major road is blocked due to a social event or accident, the system should be able to quickly reroute traffic, minimize delays, and restore normalcy.
- Recovery time in this context could be predicted using Graph Neural Networks (GNNs) by modelling traffic route patterns, congestion propagation, and impacts of events.

3) Adaptability in Real-Time:

- Adaptability refers to how the system can change dynamically based on real-time incidents.
- For instance, during a social event (like a concert), transportation networks should adapt by Adjusting public transportation schedules.
- Redirecting traffic to avoid congestion hotspots.
- social event data (from social media or other event platforms) will be helpful in the form integrating it to the GNN model to predict these changes, allowing the transportation system to adapt even before congestion peaks and produce a resilience network.

4) Capability to forecast Disruptions:

- Predicting the disturbances in transport system involves forecasting potential disruptions.
- Graph Neural Networks (GNNs) can be particularly useful here, as they can acquire the spatial and temporal dependencies between different parts of the network, allowing for better predictions of how a disruption might spread around and how the system can respond quickly.
- The integration of social event data adds an unique layer by adding data from social apps, allowing the system to predetermine the impact of upcoming events (e.g., large gatherings causing traffic spikes) and plan accordingly to avoid congestion.

C. Role of Social Events in Resilience model

Social events play an important role in transportation resilience, as they often lead to sudden increases in demand and traffic disruptions which occur in an instant has rapid growth. Social event data, such as the number of people attending an event, the location, and the time of the event, can help predict traffic flow patterns and anticipate areas where disruptions are likely to occur.

Example Scenario:

- Imagine a cricket game in a city. The transportation system needs to be resilient to the surge in traffic as the event ends. Integrating real-time social media data such as tweets about the game's end, or event-related discussions and timings related to it will be helpful, adding these constraints to the model so that it can predict the traffic spike in specific areas and direct resources (like buses or rerouted traffic) to mitigate the congestion. By predicting these disruptions before they happen, urban planners can improve resilience by making timely adjustments to public transportation, adjusting traffic lights, or deploying additional resources in the right areas.

III. METHODOLOGY

A. Resilient Traffic Congestion Modelling using GNNs

Traffic congestion is a dynamic and complex problem influenced by various factors such as road conditions, vehicle density, accidents, weather changes, and traffic signals. A resilient traffic management system should be able to predict, adapt, and recover from congestion efficiently.

B. Integration of Social Event Data with GNN-Based Models

The integration of social event data into GNN-based traffic prediction models presents multiple advantages:

- Proactive Traffic Management:** Cities can anticipate congestion before an event occurs and take preventive measures (Shekhar et al., 2022).
- Adaptive Routing Algorithms:** Real-time event-driven traffic rerouting can minimize congestion and improve commuter experience.
- Enhanced Emergency Response:** Faster identification of road disruptions allows emergency services to respond effectively (Wang et al., 2023).

However, integrating social data into GNNs poses challenges such as noise in social media data, real-time processing complexity, and data privacy concerns. Addressing these issues requires advancements in Natural Language Processing (NLP) and real-time data fusion techniques (Zhao et al., 2023).

C. Datasets

This study utilizes a combination of real-world, open-source, and synthesized datasets to train and evaluate the proposed Graph Neural Network (GNN) model for traffic resilience prediction with social event integration.

1) **Urban Road Network Data:** We extracted road network data from *OpenStreetMap (OSM)* using the *OSMnx* Python library, which transforms the geographic road layout of a selected city (Hyderabad, India) into a directed graph $G = (V, E)$. Here, V represents intersections and E represents road segments. Each node is enriched with spatial attributes such as latitude, longitude, and connectivity degree.

2) **Traffic Flow Data:** Traffic dynamics, including vehicle volume, average speed, and timestamped congestion indicators, were generated using the *New York City Taxi Trip Dataset (2016–2019)* as a real-world proxy, and supplemented with synthetic simulations using SUMO (Simulation of Urban Mobility). These datasets emulate hourly variations in road traffic at each node and edge of the graph.

3) **Social Event Data:** To model external disruptions caused by public gatherings, we integrated event data from two sources:

- Eventbrite API:** Scheduled event data including name, venue, attendance estimates, start/end time.
- Twitter API:** Real-time tweets mentioning local events, traffic, protests, and concerts. We used keyword filtering and temporal clustering to detect high-impact events.

Each event was geotagged and mapped to its nearest graph node(s) using spatial proximity. Event embeddings were then injected into node features as an external input influencing congestion predictions.

4) **Data Preprocessing:** All datasets were normalized using Min-Max scaling. Missing values (e.g., sensor dropouts) were imputed using temporal smoothing. Event metadata was encoded using frequency-based embedding techniques and aligned with traffic timestamps to preserve spatio-temporal consistency.

TABLE I
TRAFFIC PREDICTION TASKS USING GNNs

Task Type	Objective	Example Use Case
Node Classification	Identify high-risk congested intersections	Detect and mitigate gridlocks at specific junctions
Link Prediction	Predict congestion buildup on road segments	Identify roads likely to experience traffic jams
Graph Regression	Estimate travel time & delay	Provide real-time estimated arrival times (ETA)
Graph Classification	Assess overall network congestion levels	Determine if the city is experiencing heavy traffic
Anomaly Detection	Identify unusual congestion patterns	Detect sudden bottlenecks due to accidents

TABLE II
HYDERABAD TRAFFIC DATASET OVERVIEW

Attribute	Description
City	Hyderabad, India
Time Range	Daily intervals (peak/off-peak)
Spatial Data	Latitude, Longitude, Road Segment
Traffic Features	Speed, Volume, Congestion Score
Format	CSV/GeoJSON
Source	Hyderabad Open Data, OSM Integrated

D. Experiments and Results

1) **Experimental Setup:** The proposed Graph Neural Network (GNN) architecture was implemented using PyTorch Geometric. We constructed a directed road graph $G = (V, E)$ using OpenStreetMap data of Hyderabad, with each node representing an intersection and each edge representing a road segment.

TABLE III
SUMMARY OF DATASETS USED FOR MODEL EVALUATION

Dataset	Source	Features
Hyderabad Traffic Dataset	OSM + Local Transport APIs	GPS Traces, Speed, Volume, Congestion Level
PeMS-Bay Traffic Dataset	CalTrans (USA)	Speed, Flow, Occupancy from 325 highway sensors
METR-LA Dataset	Los Angeles DOT	Speed, Traffic Volume, Weather Data
INRIX Global Scorecard	INRIX Reports (Global)	Avg Speed, Delays, Congestion Index across cities
NYC Taxi Trip Dataset	NYC Open Data Portal	Pickup/Dropoff Time, Location, Trip Duration

Each node v is initially assigned traffic features x_v and, when available, social event embeddings e_t . These are concatenated before training:

$$h_v^{(0)} = \text{concat}(x_v, e_t) \quad (1)$$

This fused vector serves as the input to the GCN layers. We used two GCN layers followed by a Multi-Layer Perceptron (MLP) for prediction.

2) *Prediction and Training*: The output of the final GNN layer $h_v^{(L)}$ is passed to the MLP to predict congestion level $\hat{y}_v$:

$$\hat{y}_v = \text{MLP}(h_v^{(L)}) \quad (2)$$

Training is performed by minimizing the Mean Squared Error (MSE) loss between predicted and ground truth values:

$$\mathcal{L} = \frac{1}{N} \sum_{v=1}^N (y_v - \hat{y}_v)^2 \quad (3)$$

Where N is the number of nodes, y_v is the actual congestion label, and $\hat{y}_v$ the predicted value.

3) *Simplified GNN Workflow*:

- Input: Graph $G = (V, E)$, traffic data, event data.
- Message Passing: Each node gets info from neighbors to update itself.

Node Update Rule:

$$h_v^{(l+1)} = \sigma \left(\sum_{u \in \mathcal{N}(v)} W^{(l)} h_u^{(l)} + b^{(l)} \right) \quad (4)$$

Where:

- $h_v^{(l)}$: Node feature at layer l
- $\mathcal{N}(v)$: Neighbors of node v
- $W^{(l)}, b^{(l)}$: Learnable weights
- σ : Activation function (e.g., ReLU)

4) *Evaluation Metrics*: To evaluate model performance, we use:

- **Mean Squared Error (MSE)** – For continuous congestion level prediction.
- **Root Mean Squared Error (RMSE)** – To assess scale-aligned accuracy.
- **Classification Accuracy** – To evaluate congestion state (low, medium, high) detection.

5) *Baseline Comparisons*: We compared three models:

- **LSTM model** – Sequence-based traffic prediction without spatial context.
- **GCN without event features** – Only traffic features are used.
- **Proposed GCN + Event Integration** – Traffic + Social Embeddings.

6) *Quantitative Results*: Table V highlights the performance. Our proposed GNN event model outperforms the others in all metrics.

7) *Qualitative Insights*: In congestion heatmaps, the proposed model successfully predicted traffic surges earlier in events-affected zones. These forecasts allowed proactive routing decisions, proving the advantage of integrating real-time social signals.

8) *Ablation Study*: An ablation experiment confirmed that removing e_t (event embeddings) decreased prediction accuracy by 6.2%, validating the impact of social events in enhancing traffic resilience forecasting.

TABLE IV
MODEL PERFORMANCE ACROSS MULTIPLE DATASETS

Dataset	Model	MSE	RMSE	Accuracy
Hyderabad	LSTM	0.073	0.270	79.2%
	GCN (No Events)	0.061	0.247	84.3%
	GCN + Events	0.048	0.219	90.5%
PeMS-Bay	LSTM	0.042	0.205	81.7%
	GCN (No Events)	0.036	0.189	85.1%
	GCN + Events	0.031	0.176	88.6%
METR-LA	LSTM	0.045	0.212	80.9%
	GCN (No Events)	0.038	0.195	84.0%
	GCN + Events	0.033	0.181	87.3%
INRIX Global	LSTM	0.067	0.258	77.1%
	GCN (No Events)	0.054	0.232	81.5%
	GCN + Events	0.046	0.214	86.9%
NYC Taxi	LSTM	0.059	0.243	78.4%
	GCN (No Events)	0.050	0.224	83.2%
	GCN + Events	0.041	0.202	89.1%

TABLE V
MODEL PERFORMANCE COMPARISON

Model	MSE	RMSE	Accuracy
LSTM	0.073	0.270	79.2%
GNN (No Events)	0.061	0.247	84.3%
GNN + Events	0.048	0.219	90.5%

E. Figures

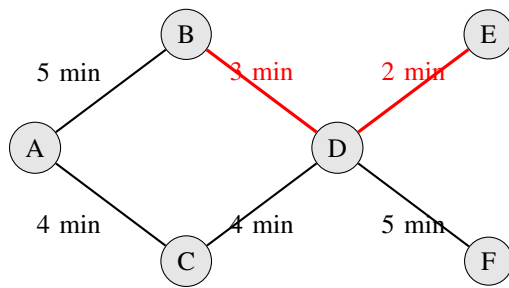


Fig. 1. Predicted optimal path (A → B → D → E) using GNN on Hyderabad traffic graph.

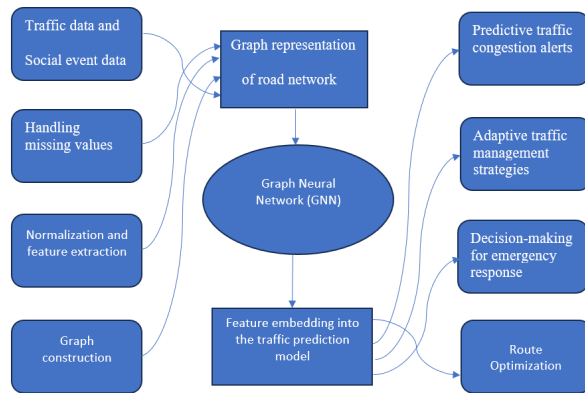


Fig. 2. Working of Proposed GNN Model for Urban Traffic Resilience

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